**New Models of Interdiction in Networked Systems**

*Alexander Gutfraind, Los Alamos National Laboratory, agutfraind.research@gmail.com*

## Introduction

Sixty-six years ago, in 1955, General Frank Ross, formerly in charge of the US Army's Transportation Corps in Europe, commissioned RAND analyst Ted Harris to solve a problem. General Ross wanted a plan for cutting the Warsaw Pact rail network in case of a hot war in Europe (Schrijver, 2002). In that network, which rail nodes would need to be bombed in order to disrupt Soviet supply routes? Every possible route from the origin in the east to the battlefront in the west would have to be disrupted. Obviously, some railway lines have more capacity than others and traffic can be re-routed from damaged lines to functioning ones. Which one of the many target sets is best?

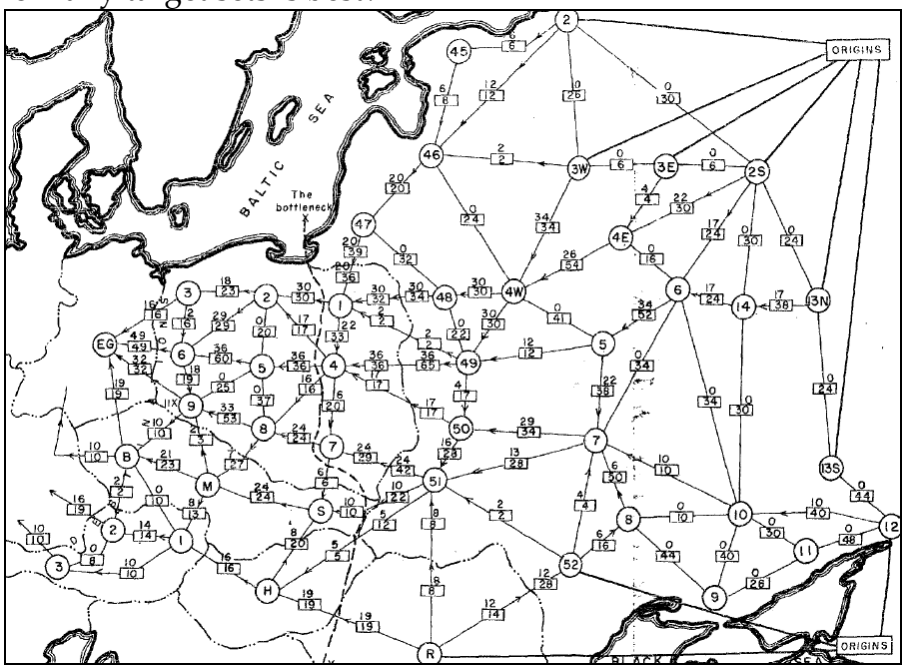



**Figure 1: The East European railway network and its bottleneck, the minimal cut,** from Harris and Ross (1955). Circles are labeled nodes and boxes give edge capacities (number of 1000-ton trains per day). Numbers over boxes give the flow when the network is fully utilized as during a war.

In network language, this problem is now called the "minimum cut problem": a railroad hub is represented by a node which is connected to nearby nodes using edges of some specified carrying capacities. The capacities correspond to the amount of cargo they can ship and also roughly to the difficulty of destroying them. The objective is to destroy some of those links so as to fully disconnect the supply nodes of the adversary from his targets. So, it was a monumental scientific achievement when Ford and Fulkerson pioneered methods for solving this problem optimally even on very large networks.

Ross' minimal cut problem was unusual in '55, but is typical of today's battlefield. While traditional battles were fought on land, sea, and air, more and more the conflicts are set in networks: road and air transportation systems, clandestine networks, and most recently cyber networks. While in the past the objective was to protect your own territory or to take the enemy's, today's objective is protecting your own networks while unwiring the enemy's. In the past, commanders relied on cartographers and logistics specialists, but today's strategists must also rely on network scientists. In this article, I survey recent development in this field, or to be specific, the research presented at a session of the 2010 INFORMS annual meeting.

**The Connected Minimal Cut**

Not every network disruption problem is like Ross' minimal cut. In a minimal cut, the disrupted edges can be far removed from each other, creating a cut through a subtle synergy. In practice, such distributed attacks might be infeasible and it is necessary that the disrupted edges are close to each other. For example, in the case of air strikes, to make any arbitrary cut one might need to penetrate enemy air defense systems in multiple locations far from each other. Far more practical is to suppress those defenses in part of the airspace and then disrupt the network in that area. A similar penetration problem is also relevant to cyber-attacks. Finally, one can consider the social terrain: in disrupting clandestine networks such as terrorist groups, it is very difficult to cut the network by arresting enough individuals because many of the actors are difficult to access. A more realistic disruption plan is to infiltrate the network and then disrupt or even recruit other nodes *connected* to the initial infiltration set (Fig. 2).

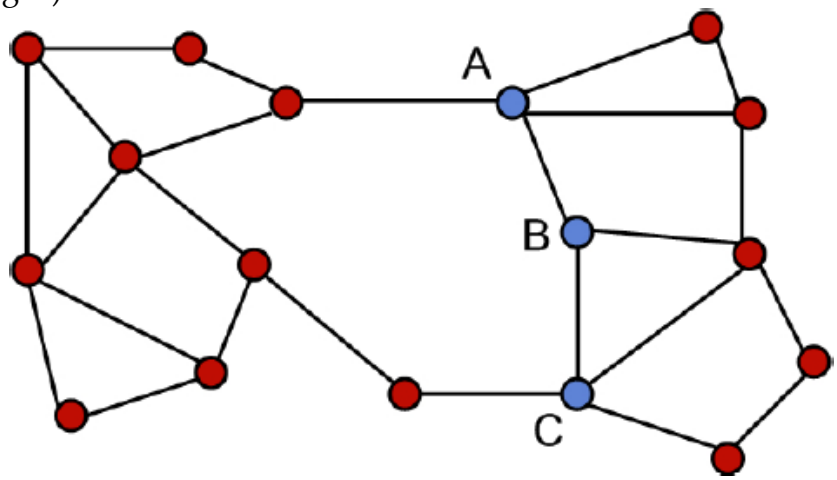



**Figure 2: A Connected Cut is formed by nodes A, B, and C.** The network could be infiltrated at A and C, then disrupted at B. When the three are removed, the left part of the network is separated from the right. Adapted from Banerjee et al. (2011).

The work of Banerjee and collaborators considers the *minimal connected cut* problem (MCC), where the network G is to be cut with minimal effort while requiring that the removed set forms a connected subgraph of G. Can such a subgraph always be found? Sadly, no. MCC is shown by the authors to be computationally hard, NP-hard (for the

enthusiasts, MCC could be related to the Steiner Tree problem - the problem of building with the least cost, a spanning network between a specified set of nodes.) NP-hardness indicates that it would be computationally difficult to find the MCC on *arbitrary networks*. Fortunately, it is often possible to quickly find solutions that are good enough, if not quite optimal, even on large networks. Moreover, real-life networks are not truly arbitrary and often have a special structure. Indeed Banerjee *et al* found an important special case: networks that sit on a plane (without crossing edges). In those planar networks, MCC is easy to find. This is great news because many infrastructure networks are planar or nearly so. Cyber networks as well as social networks are usually not planar: if 5 nodes are mutually connected, the graph is no longer planar. Therefore, an obvious future project is to develop algorithms that find high-quality solutions to MCC even on non-planar networks.

**Tactical Network Interdiction**

The problems considered by General Ross and Ted Harris are sensible for a total war: all the enemy's supply lines must be cut whatever the cost or collateral damage. In a limited war (or with a smaller attack force), a full network cut is not possible, and one may instead try to disrupt some of the network. Such a poor man's cut is the Network Interdiction (NI) problem (Israeli and Wood, 2002). In NI, success is measured by how much more difficult it becomes to cross the network.

To give a concrete example of recent interest, imagine that the enemy is a nuclear-armed terrorist who starts somewhere on the network and wishes to reach a destination. One might be able deploy radiation sensors which detect the materials inside such a weapon, but the sensors are only feasible in some parts of the network such as border crossings. Those sensors increase the risk to the adversary and may force him to find long risky routes that bypass the sensors (see Morton et al. 2007.)

Recently, Mehmet Ertem and Vicki M. Bier from the University of Wisconsin-Madison applied Network Interdiction ideas to defending against cyber attacks. In their model the network represents the "attack graph" – the set of all possible attack scenarios against the system. An attack scenario is just a path on the network that reaches from an entry point to another node, for example, secure data stored deep in the network. Each of the edges in the network has an associated success probability when traversed by the attacker. At this point, the defender deploys sensors that further reduce the probability of success. He may also install systems such as firewalls, which fully block suspected malicious traffic. The attacker and defender play a multi-step game where the defender can learn from past attacks to add new sensors to the network. This problem could be represented as a multi-stage stochastic program. Because programs of this type are hard to solve optimally, Ertem and Bier propose a number of heuristics that help find effective interdiction solutions.

**Network Interdiction of Transnational Terrorist Networks**

Much of the recent work in network interdiction was motivated by problems in counter-terrorism. Indeed, terrorist operations rely on networks of many kinds: the internal

organization of the terrorist group is a network, and a network may represent the travel routes towards the target.

A terrorist operation can itself be represented as a network with interdependent steps: the recruitment of operatives, the procurement of weapons, training with the weapons, and so forth. An important additional problem for a transnational terrorist organization like Al-Qaida or Hizbullah is the risk of crossing international borders: getting the travel documents, learning the language, and the logistics at the remote target. They must also consider the efficiency of law enforcement agencies and the value of different targets - countries.

One could reconstruct the calculus of the terrorists, that is, rebuild their operational network (Fig. 3). In that network, much of the relevant information could be estimated from public sources (if only roughly). For example, the costs of crossing international borders are indirectly expressed in the amount of tourism or immigration between the relevant countries (properly adjusting for populations and distance.)

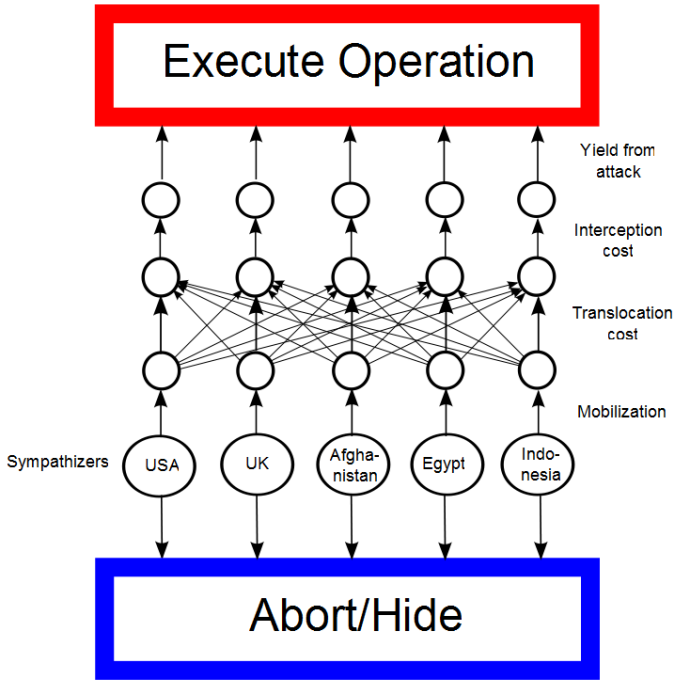



**Figure 3: A small operational graph for transnational terrorists.** Rows represent stages while columns represent countries. Every attack corresponds to a path from a sympathizer node in some country to the column of the country being attacked, and then to the "Execute Operation" node. The full model contains many more source and target nations.

After collecting such data, one can estimate the risk to different countries – the likelihood of an attack. To be more specific, suppose $T_{ij}$ represents the risk for moving a cell from country i to country j. Also, let $I_j$ be the risk of interception once at j, and $Y_j$ the yield from a successful attack. Note that risks or probabilities can be converted to cost by the function $f(x) = -\log x$, which make them comparable to the yield. Then "Bin Laden's"

problem is to find which country j to target:

$$\min_j \left( T_{ij} + I_j + Y_j \right)$$

To find the total risk to any country one must also consider the supply of cells in various source countries (which could be estimated from public opinion surveys and other data). The final estimate is as follows. The United States has the highest risk of any country from international terrorism. This is simply because by attacking the US, a terrorist organization hopes to affect the policy of the biggest player in the international arena. Of course, in the last decade the US did not stand idle and made it much more difficult for terrorists to reach its shores. So, a possible scenario in the network is one where all edges entering the US are interdicted. In response, smart terrorist groups can be expected to shift to less protected but still valuable targets. Under the assumptions of the model, such a US move would not measurably decrease the volume of attacks but would deflect them, greatly increasing the risk from terrorism to every other country.

The model is consistent with the increase in terrorism in Europe following the security measures implemented over the last decade: Islamist radicals based in Europe or going through Europe just could no longer reach the US! This effect points to the need for international security arrangements, because in the current environment much of the security effort merely fuels a competition over which country is hardest to reach or has the most fortified embassies. In principle, a more effective strategy is to focus resources on stopping terrorists at their source nodes. In practice, such interdiction strategies might have unwanted effects of their own.

Weighing the costs and benefits of interdiction strategies and computing the vulnerabilities and resiliences of networks is the task of network science. It is certain that this research area will only increase in significance as the world grows to become more and more a cake of overlapping networks.

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**Author Biography**

Alexander Gutfraind received a Ph.D. from Cornell University, and is currently a postdoctoral researcher at the Los Alamos National Laboratory. He develops mathematical models to illuminate problems in complex systems and counter-terrorism using methods from the theories of complex networks, discrete optimization and dynamical systems.

